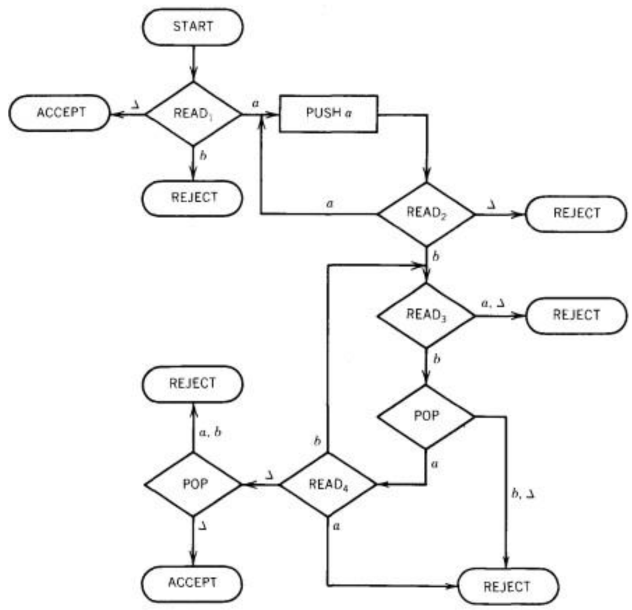
Theory of Automata

Assignment No. 4

Submit the Hard copy to your Instructor's office (or In class) before 5:00 pm on 4th December.

1. Consider the following PDA.



A. Trace the following words in PDA. Also show what happens to INPUT TAPE and STACK as each of the following words proceeds through the machine.

   a) abab
   b) aaabaa
   c) abb
   d) aaabbbb

B. Find a CFG that generates this language ?

C. Is this language regular?

2. Make a PDA that accepts this language

$$S \rightarrow Xa \mid Yb$$
$$X \rightarrow Sb \mid b$$
$$Y \rightarrow Sa \mid a$$

3. Design PDA for the language:
   1. $a^n b^n c^m d^m e^p f^p$
   2. $a^n b^m c^p e^m f^n$
   3. PARENTHESIS = $\{ \Delta, (), (()), ((())), (())(), ()(()), ()()() \}$
4. Draw 2PDA of DOUBLE FACTORIAL = $\{ a^{n!} b^{n!}$ where $n>0 \}$
5. Draw TM for ODD-ODD (number of a's and b's are odd).
6. Build a TM to accept the language $\{a^n b^n a^n\}$ based on the following algorithm.
   I. Check that the input is in the form a*b*a*
   II. Use DELETE in an intelligent way.
7. Design a Turing machine that takes input a positive number x and performs the computable function $f(x) = x^2$. Assume the input is in unary notation, the tape head points in the start of the input. However, you may leave the tape head at any location on the output string when the computation is done. You are also allowed to use the sub programs of INSERT and DELETE. You may assume after INSERT operation, tape head points at the newly added cell while after DELETE operation, tape head points at the same location. An example is given below for your understanding:

Status of tape on input:

#	1	1	1	Δ	Δ	.	.	.

Status of the tape at output:

#	1	1	1	1	1	1	1	1	1	Δ	.